



OPTIMIZATION OF AN SRF GUN DESIGN FOR UEM APPLICATIONS

Ao Liu,
Roman Kostin, Chunguang Jing, Pavel Avrakhov,
Euclid Techlabs and Euclid Beamlabs,
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Outline

- Introductions



- Design Concept



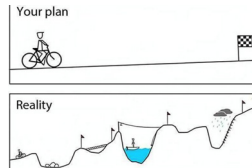
- Technical details



- Simulations



- Design Optimization



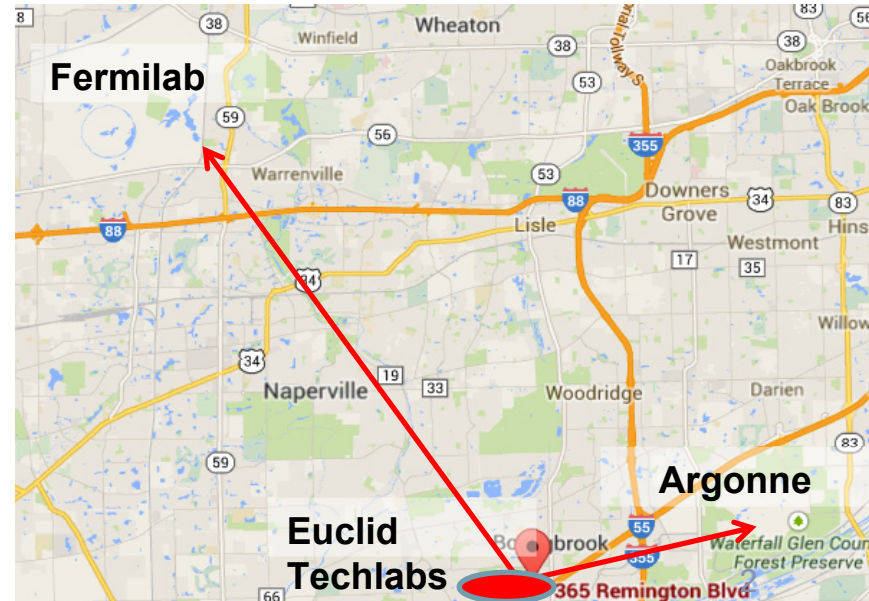
- Summary

Project funded by DOE under **Phase II** Grant #DE-SC0018621

Introductions to Euclid

- Euclid Techlabs, LLC, founded in 1999 is a company specializing in the development of advanced high power/high frequency devices and beam instrumentation for various applications. Areas of expertise include electron accelerators, photoinjectors, dielectric accelerators, pulsed electron microscopy, accelerator simulations and algorithm development, new material technology, photocathodes, etc.
- www.euclidtechlabs.com. Find us on Facebook and LinkedIn!
- Two offices: administrative office in Washington, DC; research and lab facilities in Bolingbrook, IL
- Tight collaborations with national labs and universities: FNAL, ANL, Jlab, BNL, NIST, NIU, IIT, CERN.

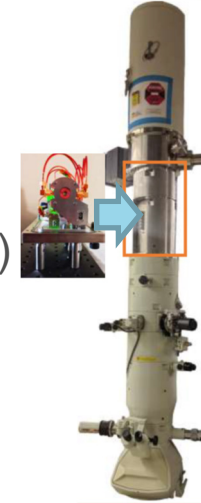
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Euclid Research, Technologies and Services



- Compact electron accelerator and its test facility (bunker)
 - 1 MeV accelerators (**THYBB3**, S. Kuzikov, oral talk; **WEPLM21**, **WEPLO11**, **TUPLH07**)
 - SRF photogun (this talk)
- Beam instrumentation
 - Electron beam profile monitor (**TUPLE04**, A. Liu)
 - X-ray or charged particle detector (**TUPLH08**, S. Antipov)
 - Black gun for DC photoinjectors (**MOPLO06**, E. Montgomery)
- Laser technologies
 - fs laser microfabrications (**MOPLH16**, S. Antipov)
- Photocathodes
 - Nanostructure photocathode for spin-polarized e⁻ beam (**MOPLH11**, E. Montgomery)
 - UNCD photocathodes / FE (**MOPLH06**, G. Chen)



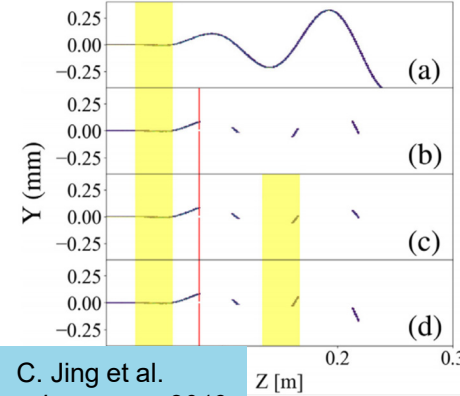
Others:

- Tunable GHz TEM pulser: published @ *Ultramicroscopy*
- Magnetron sputtering: for Fermilab Mu2e production target and more.
- RF beam instrumentations

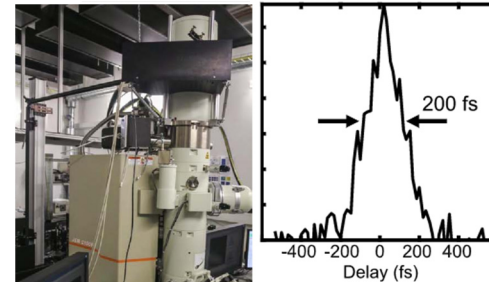
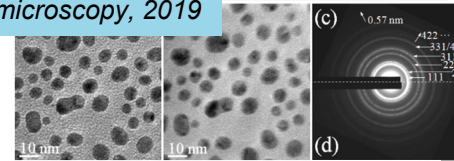


Dynamic, fast TEM: new era of dynamic microscopy

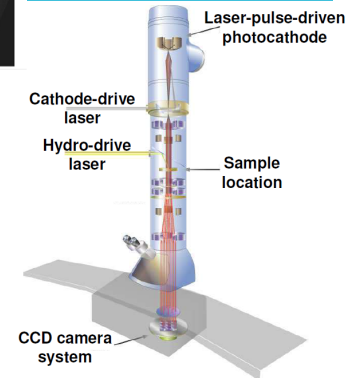
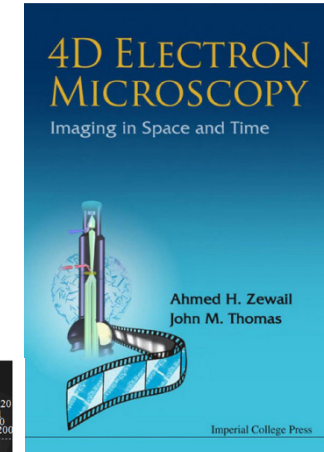
- Pump-probe experiments using synced RF or laser pulses
- Hundreds of fs or ps electron pulses achieved by modifying a TEM
- Pulsed stroboscopic TEM: chopping DC beam to achieve GHz-repetition rate, ps-long electron bunches
 - Enabling dynamic structure microscopy using a regular TEM machine
- Single-shot laser-driven photocathode-based TEM
- E_{kin} at 200-300 keV, strong space charge effect at the high e^- density sample area



C. Jing et al.
Ultramicroscopy, 2019

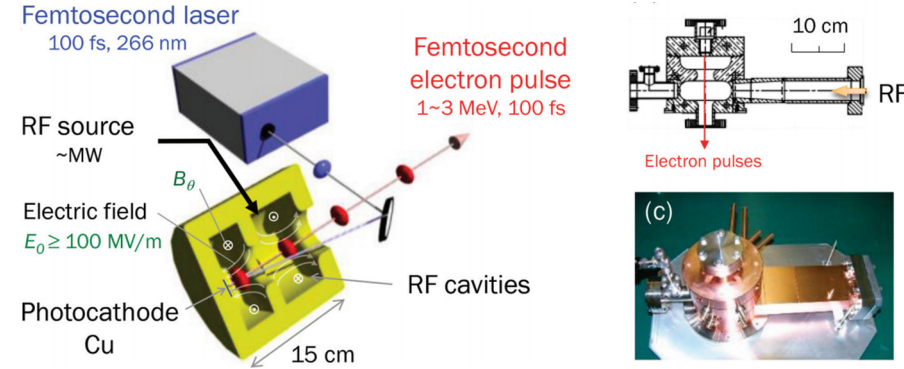


A. Feist et al.
Ultramicroscopy, 2019



MeV UEM/UED: adding ultra- to fast, adding -er to small

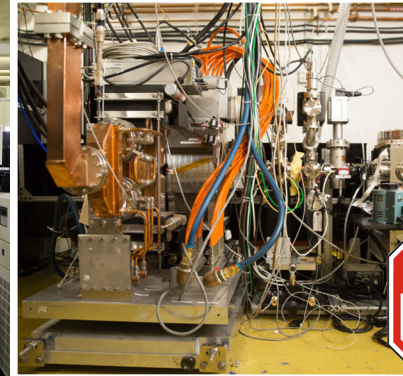
- MeV beam greatly suppresses the space charge effect at high bunch charges
- Stringent requirements on energy spread $\frac{dE}{E}$ and bunch length σ_t
 - UED: $\frac{dE}{E} \sim 10^{-4}$, $\sigma_t \sim$ tens of fs, less than 500 fC
 - UEM: $\frac{dE}{E} \sim 10^{-5}$, $\sigma_t \sim$ ps to ns, 500 fC or more
- Room temperature photogun (CuRF) (e.g. at Osaka University, Japan) provides 3 MeV $\sim$ pC in 100 fs
 - Large facility needed + high RF power consumption
 - RF jitter ~ 100 fs;



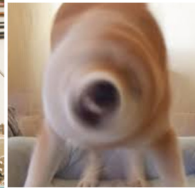
J. Yang, Electron microscopy



UEM @ Osaka U

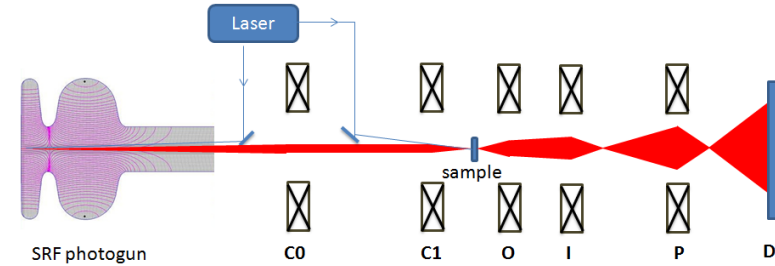


UED @ SLAC



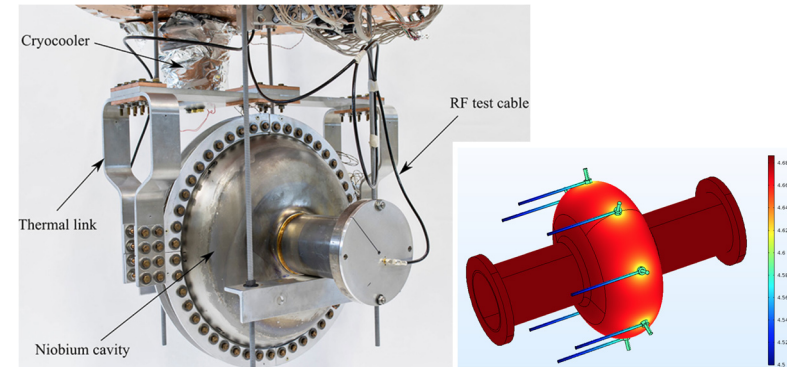
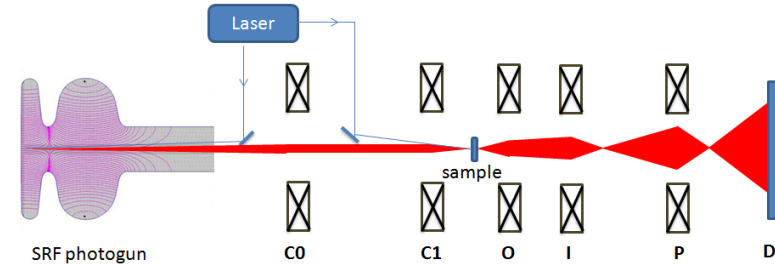
SRF photogun MeV UEM/UED: It's time

- **Q:** Why SRF? **A:** Accurate control of both RF amplitude and phase, CW operation
- **Q:** Technical details? **A:** 1.3 GHz, 1.X cell SRF cavity, use back wall as the photocathode, ~ 2 MeV beam, low level RF (LLRF) control
- **Q:** What's **shiny**? **A:** Novel conduction cooling, Nb₃Sn coating. Benefits: The entire system cooled by a commercially available crycooler free of liquid He; Less vibration from a full cryosystem.



SRF photogun MeV UEM/UED: It's time

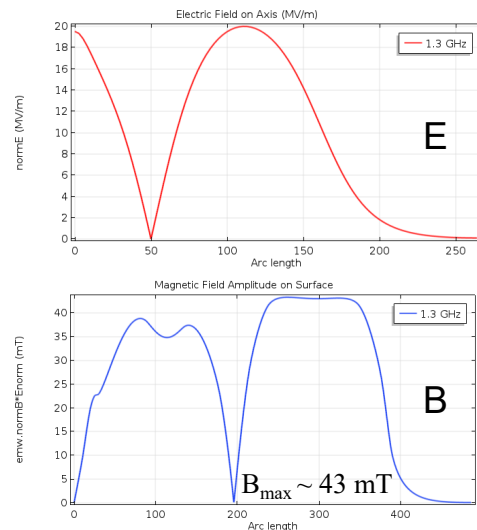
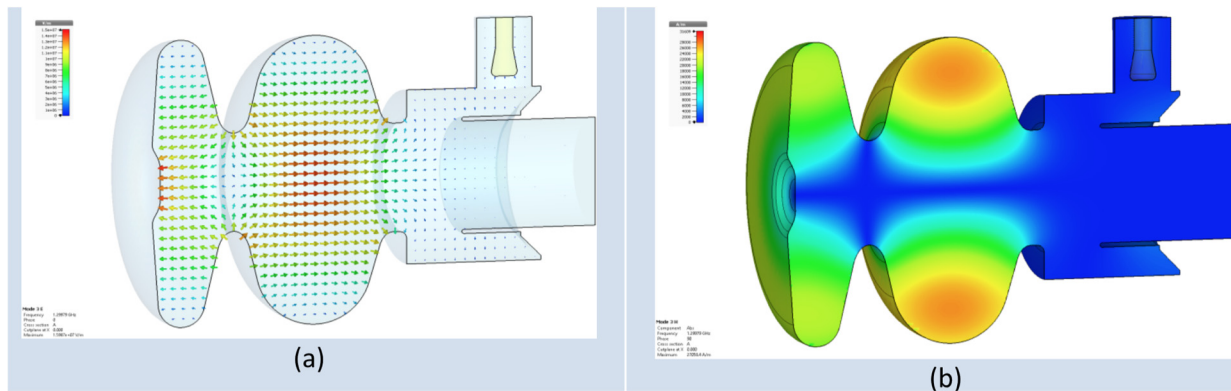
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- **Q:** What's **shiny**? **A:** Novel conduction cooling, Nb₃Sn coating. Benefits: The entire system cooled by a commercially available cryocooler free of liquid He; Less vibration from a full cryosystem.
- **Q:** Where will it be tested? **A:** Fermilab or Cornell are the most probable hosts for now



Courtesy of Fermilab, also see IEEE TRANSACTIONS ON APPLIED SUPERCONDUCTIVITY, VOL. 29, NO. 5, 2019 CC cavity demonstrated at Fermilab; Euclid provided LLRF

Cavity design

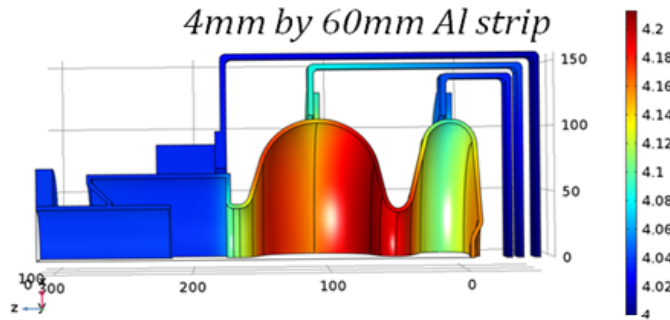
- Determine cavity length: syncing the phase for highest E_{kin} and that for the lowest $\frac{dE}{E}$
 - 1.X cell, find X!
 - Back wall is better to have an “inward slope”, adding mechanical strength.
- Proposed a “nose” structure for the back wall to fulfill the tasks above;
- Manually tuned the nose for many iterations to reach the first version
- Does it work? Yes!



- Q at 4°K: $\sim 1 \times 10^{10}$, estimated RF power dissipated: 2 W, within cryocooler capability.

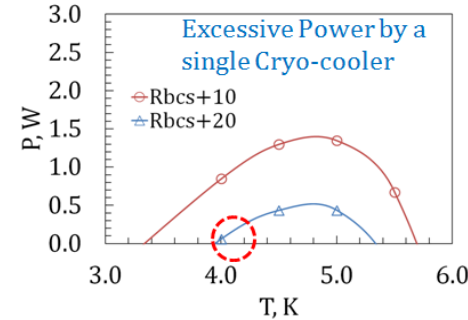
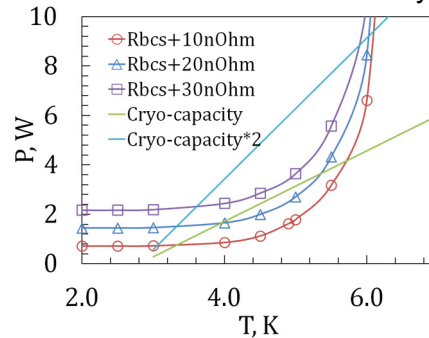
Thermal analysis of the photogun

- Analysis done with assumption of gradient at 20 MV/m



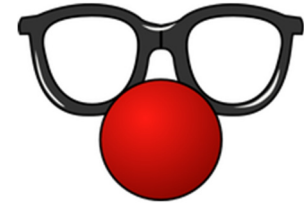
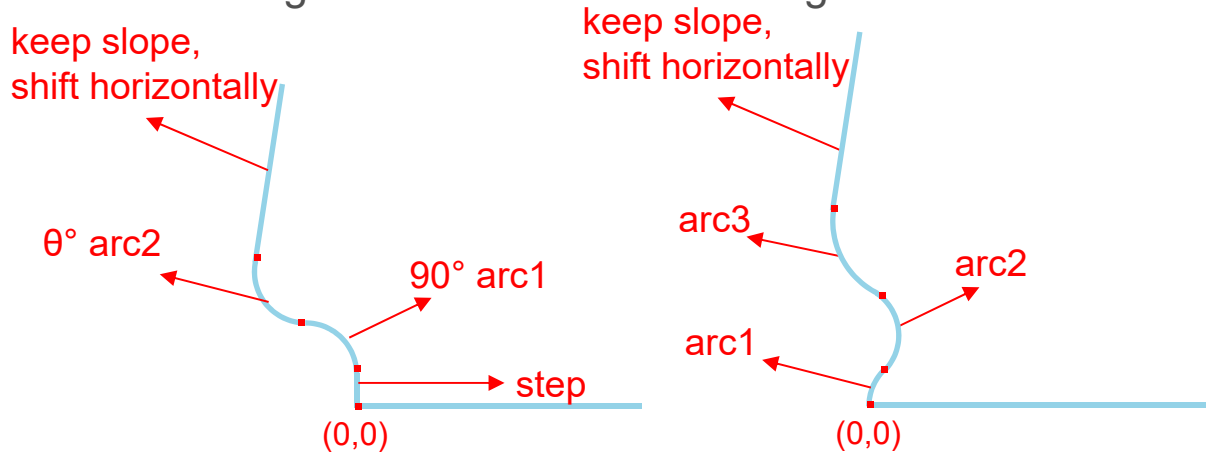
- 2 W cryocooler at 4.2 °K assumed
- If running at more than 20 MV/m field gradient, more than one cryocooler capacity is needed

Courtesy: R. Kostin



Nose optimization

- New scenarios were proposed for the back wall nose design;
 - Keeps the inward slope for now assuming same mechanical strength requirement
- Multivariate, multi-objective optimizations;
 - 4 and 5 variables for the two scenarios, respectively;
 - Each solution needs to be evaluated by first E&M simulations, then multi-particle tracking
 - Constraints to add geometric limits to the designs: curvature closes and continuous



pyaopt: a Python optimization package

- Stands for **P**ython-based **a**dvanced **o**ptimization suite
- Supports popular Multi-Objective (**MO**) and Single-Objective (**SO**) algorithms, such as the Genetic Algorithm (GA) and Simulated Annealing (SA), etc.
- Runs on supercomputers like NERSC
- Has been tested and applied to many tasks in the HEP experiments and particle beamline designs, e.g. nuSTORM, MICE



Big but friendly



Aggressive but small

Tibetan mastiff



Survives well, but not always. What if
their genes crossover?

Many
generations

Optimization using pyaopt

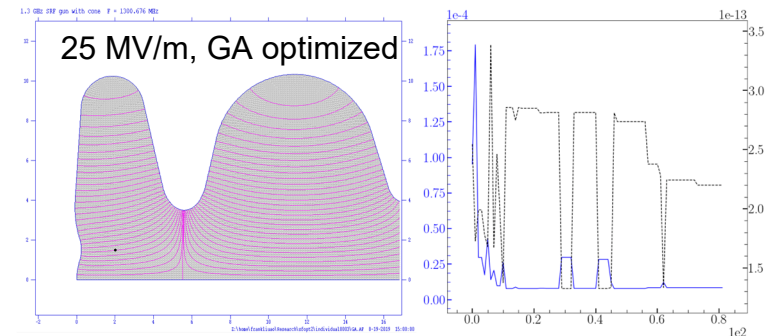
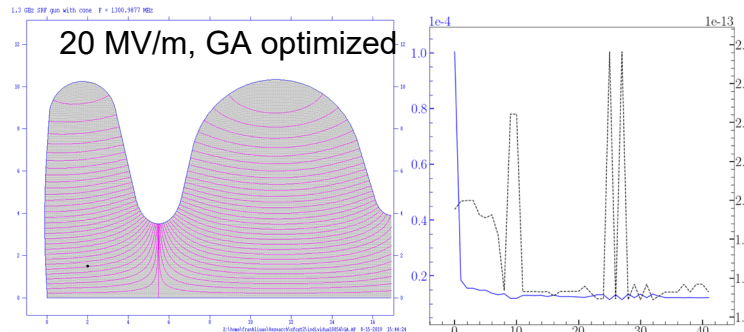
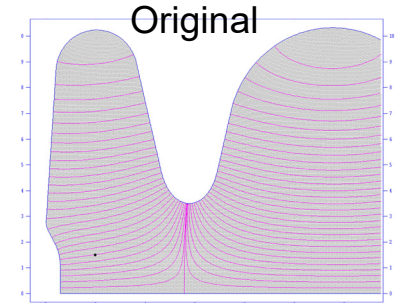
- Parallel cases (individuals) run accomplished on Ubuntu server

- Superfish (Wine) + Astra combined execution;
- X-windows is still a problem!



- GA used in this study

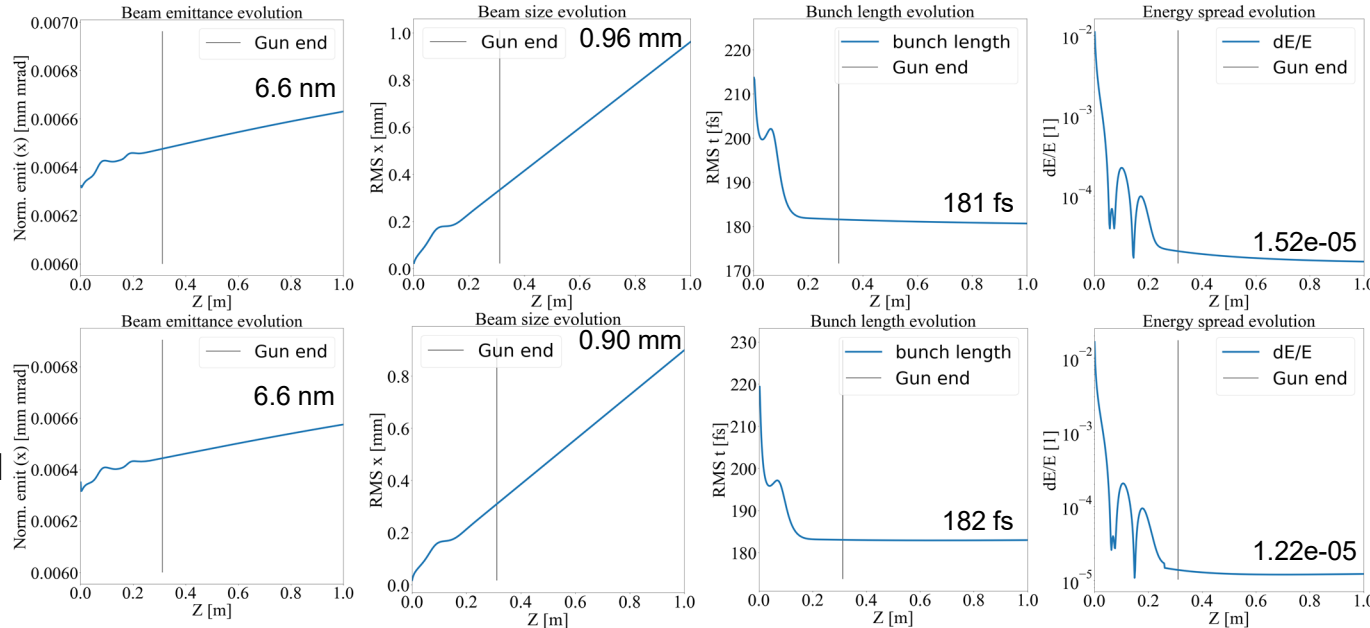
- 5 fC, 15 fs FWHM, 36 μm diameter initial beam;
- Scenario 2 outperforms scenario 1;
- Also tried to raise the maximum E_z to 25 MV/m – OK in terms of the structure, but need more cooling;



Comparison with the original case: 20 MV/m

- 5 fC, 15 fs FWHM, 36 μm diameter

Original

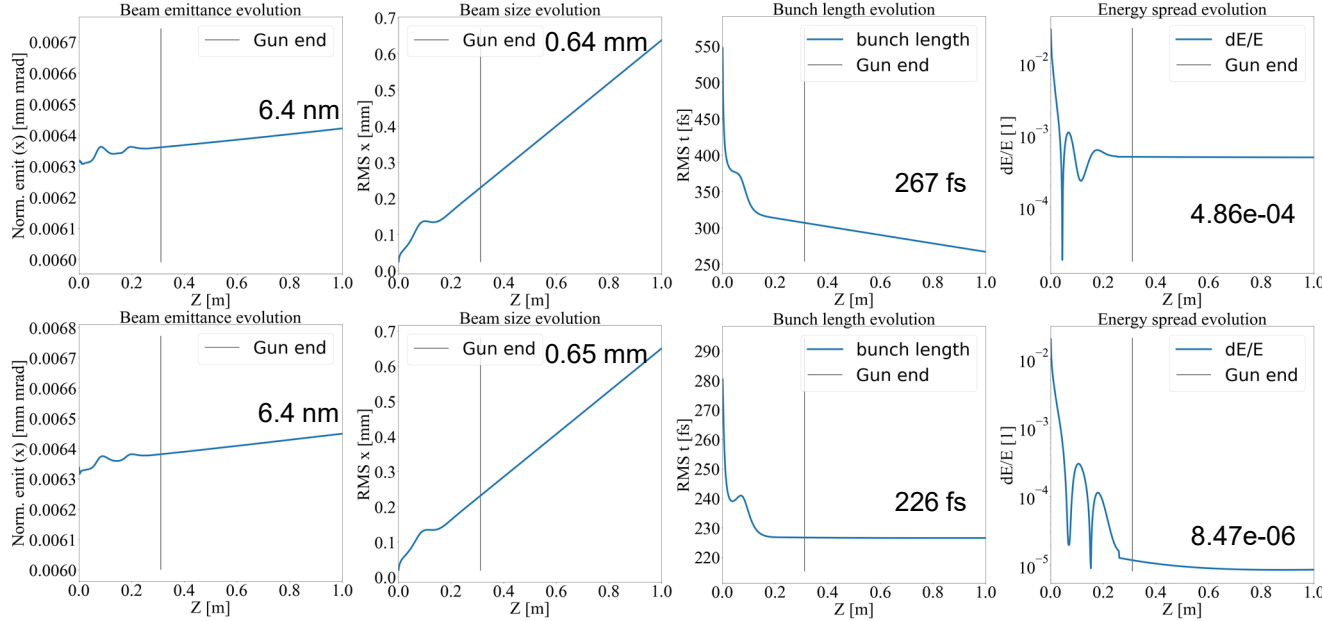


- pyaopt was able to not only find the back wall nose shape, but also the correct emission RF phase (considering optimal phase for different shapes varies)
- Slight improvement from the original manually tuned nose: probably limited by inward slope
- Proof-of-principle for further directly beam-based optimizations

Comparison with the original case: 25 MV/m

- 5 fC, 15 fs FWHM, 36 μm diameter

Original

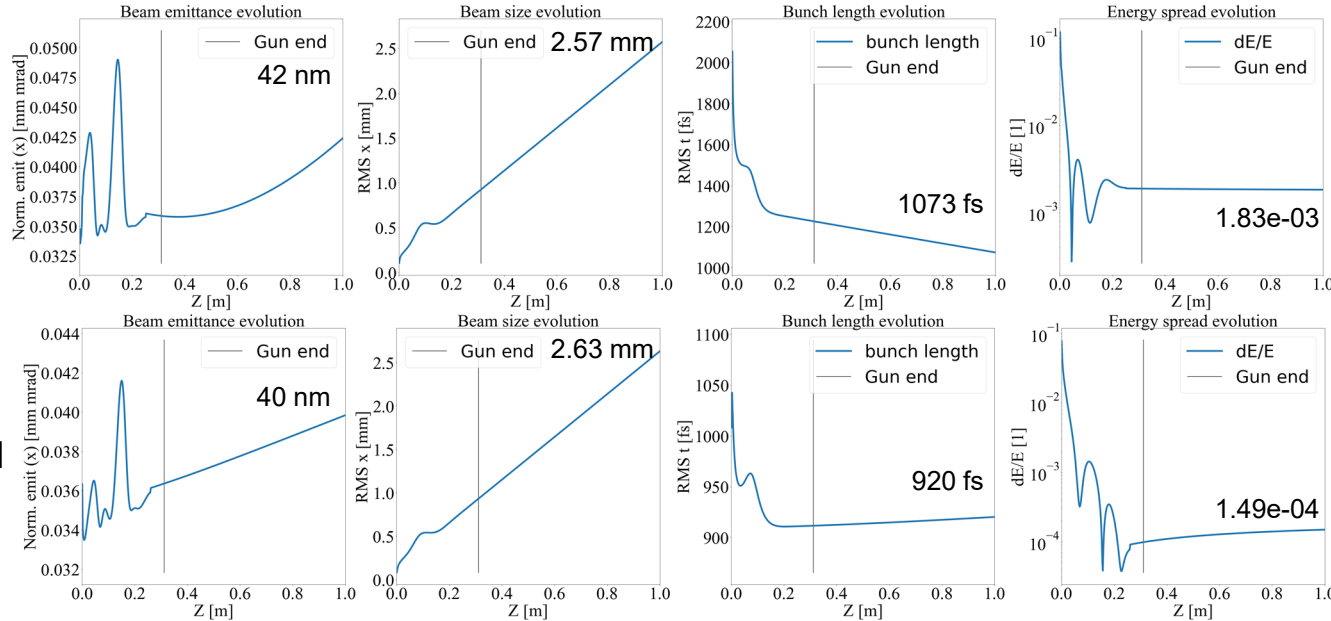


- Optimized shape for different maximum E_z is also different (obviously...)
- All parameters will be combined in the final decision being made soon.

Comparison with the original case: 25 MV/m (Cont'd)

- 500 fC, 70 fs FWHM, 180 μm diameter

Original



GA-optimized

- Rerunning the optimization is more critical for stronger E field
- Fast-turnaround runs with pyaopt opens up the opportunity to see and try more!

Conclusions

- Euclid is designing and building an MeV SRF photogun running at ~ 1.3 GHz
- Using the novel conductive cooling and Nb_3Sn coating, the system can be cooled by a commercial cryocooler, thus dramatically reducing the operation complexity and cost
- The final touches on the SRF cavity back wall design are being done. Using *pyaopt*, quick turnaround optimizations can be done with more and more realistic constraints (cooling power, mechanical strength, coating, etc.)
- The project is under Phase II SBIR funding. The module will start being fabricated very soon
- **Expect an SRF UEM from us!** Contact aoliu@euclidtechlabs.com for more discussions.

Future work and acknowledgements

- SRF photogun:
 - Finalize the nose design and optimization
 - Determine host, fabricate the 1.X-cell cavity
- *pyaopt*:
 - Implement pyCUDA / Google's JAX for more parallelization options
 - Implement algorithms suitable for online optimizations, such as the RCDS algorithm [X. Huang et al.]
- Joint optimization using SuperFish and Astra:
 - Implement on HPCs: how?
- Thanks to our collaborators, and Euclid colleagues: C. Jing, R. Kostin and P. Avrakhov, S. Ross, for the inputs on SRF cavity and system design.

Thank you!

